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# Melbourne Research Cloud

<https://docs.cloud.unimelb.edu.au/>



# Interacting with OpenStack Cloud

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- Horizon Dashboard
- OpenStack Command Line Interface (CLi)
  - *nova, neutron, cinder, glance ...*
- OpenStack API
- OpenStack SDK
  - OpenStack SDK
  - Boto
- Infrastructure as Code (IaC) Tools
  - Terraform
  - Ansible
- OpenStack Client (OSC)
  - *openstack server /network / volume / image / coe ...*



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# MRC Workshop - Login

Login to the Dashboard (<https://dashboard.rc.nectar.org.au/>)

ARDC  
Nectar  
Research Cloud

Dashboard

Support Centre Services Tutorials Request Help yao.pan@unimelb.edu.au

SELECT PROJECT  
unimelb-comp90024-...

Learn more about allocations

RESOURCE BUNDLE  
Custom

PROJECT USAGE  
1.65 / Unlimited SU

Usage Summary | Limit Summary

Home

Project

Identity

Applications

Allocations

Good afternoon Alwyn

Queensland Zone compute node upgrades [View Announcement](#)

Welcome to the ARDC Nectar Research Cloud

The ARDC Nectar Research Cloud  
Australia's National Research Cloud,  
providing computing infrastructure to researchers.

Watch on YouTube

Watch Later

Share

Welcome to the Dashboard!

The Dashboard is your web interface to the ARDC Nectar Research Cloud - Australia's national research cloud, providing computing infrastructure to researchers, giving you access to powerful computing resources without the need to purchase or host your own hardware.

New to Nectar? Don't worry, we've got loads of support articles and tutorials to help you along the way. You can learn all about the Nectar Cloud and The Dashboard on the Nectar Support Site and if you need help, you can simply contact our Support team by creating a help request.

[Support Centre](#) [Tutorials](#) [Live Training](#)

To get started with the Nectar Cloud, we recommend following the [Cloud Starter Series](#) of tutorials.

Resources for Researchers

Accelerate your research with our curated resources for researchers. We have all your digital research needs covered — datasets, tools, skills and more.

[Learn more](#)

QUICK LINKS

Here are some ways to get started:

Learn

[Tutorial: Cloud Starter Series](#)  
[Learn about allocations](#)

Actions

[Launch an instance](#)  
[Create a reservation](#)  
[Deploy an application](#)

View

[Compute Overview](#)  
[Usage Trend](#)  
[Allocation Usage](#)

Support

[Nectar Support Centre](#)  
[Browse Tutorials](#)  
[Nectar System Status](#)

USAGE

You have used

2.31

Service Units in the past 3 months.

[Help](#)

To view more usage data, visit the [Usage Trend](#) page.



- **Project**

Organizational units in the cloud (aka. Tenants). Each user is a member of one or more projects. Within a project, a user creates and manages resources.

- **Compute**

- **Overview**

- View resource allocation and usage

- **Instances**

- View, launch, and manage instances, or connect to them through VNC.

- **Images**

- View and manage images and instance snapshots created by project users.

- **Key Pairs**

- View, create, edit, import, and delete key pairs.



# MRC Workshop – Horizon Dashboard

- Key Pairs

A keypair is a pair of files used to allow access and encrypt communications. Each pair consists of a public and a private file. The public file is shared freely with the world. The private file should be locked securely, much like a password.

- How to Create a Key Pair

- Create via the dashboard

- Create a key pair locally than import the public key

- *ssh-keygen -t rsa -f mykeypair*

- *chmod 600 mykeypair* (or get UNPROTECTED PRIVATE KEY FILE! error)

```
> ssh -i mykeypair.pem ubuntu@172.26.133.125
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
@                WARNING: UNPROTECTED PRIVATE KEY FILE!          @
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
Permissions 0777 for 'mykeypair.pem' are too open.
It is required that your private key files are NOT accessible by others.
This private key will be ignored.
Load key "mykeypair.pem": bad permissions
ubuntu@172.26.133.125: Permission denied (publickey).
```

- Share a private key? NOPE! 



# MRC Workshop – Horizon Dashboard

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- Volumes
  - Volumes  
View, create, edit, and delete volumes.
  - Backups  
View, create, edit, and delete volume backups.
  - Snapshots  
View, create, edit, and delete volume snapshots.
- Cinder Backups vs Cinder Snapshots
- Volume multi-attach  
<https://docs.openstack.org/cinder/latest/admin/volume-multiattach.html>
- Volume expansion



# MRC Workshop – Horizon Dashboard

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- Network
  - Network Topology  
View the network topology.
  - Networks  
Create and manage public and private networks.
  - Routers  
Create and manage routers.
  - Security Groups  
View, create, edit, and delete security groups and security group rules.
  - Floating IPs  
Allocate an IP address to or release it from a project.



# MRC Workshop – Launching a new VM

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- Details

- Instance Name

- Name of the instance and the initial host name. Less than 63 characters.

- Description (optional)

- Brief description of the virtual instance.

- Availability Zone

- Only one AZ on MRC (*melbourne-qh2-uom*).

- Count

- Possible to launch more than one instance at once.

- Source

- Select Boot Source

- Use “image” as the Boot Source

- Image Name

- Any Linux image of your choice. Recommend: NeCTAR Ubuntu 22.04 LTS (Jammy) amd64





# MRC Workshop – Launching a new VM

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- Flavor

Defines the compute, memory, and storage capacity. Choose the one suit your needs and within the resource quota.

- Network

- Use the default Network.
- A private IP will be assigned to your instance. Need to connect to [UniMelb VPN](#) if visit from off-campus.
- You can add multiple networks to an instance (Not in this project!).

- Security Groups

Use the default security group, which allows connect to SSH port (port 22) from anywhere.



# MRC Workshop – Launching a new VM

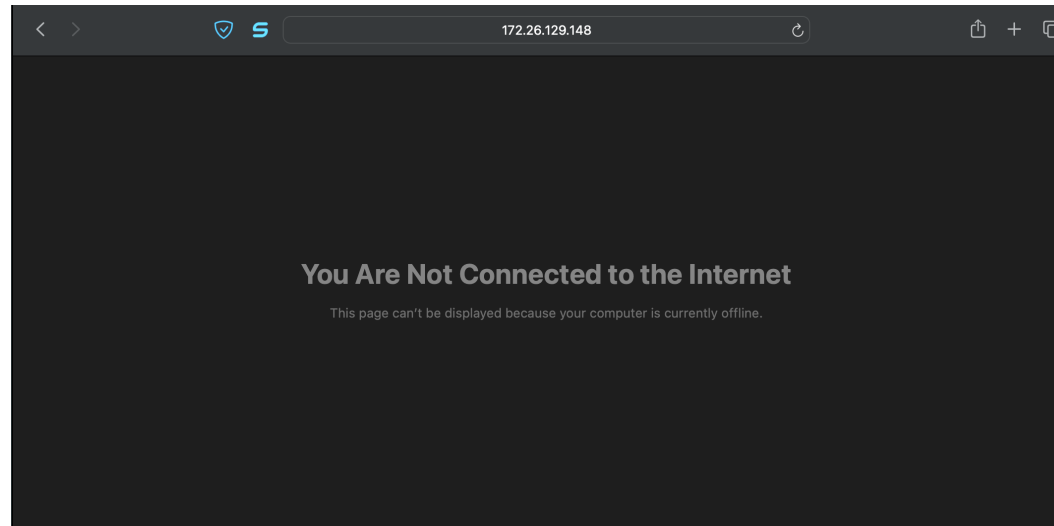
---

- Key Pair
  - !! Important !! Must select a key pair or you won't be able to access the instance!
- Connecting to VM via SSH (Secure Shell)
  - Private key
  - `ssh -i <private-key> <username>@<hostname>`
    - E.g., `ssh -i mykeypair ubuntu@example.com`
- Install a web server (Nginx / Apache)
  - `sudo apt update && sudo apt install -y apache2`  
`sudo` = runs commands with security privileges of another user (by default: superuser)
  - `curl localhost`  
use `curl` to confirm the Apache server is running

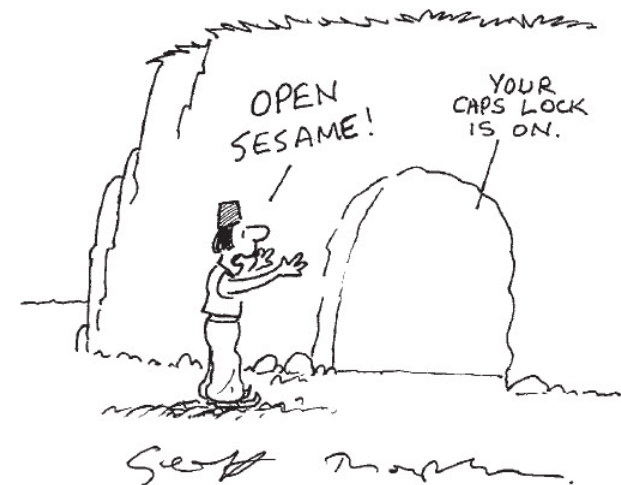


# MRC Workshop – Security Group

- Access the web server from browser



- The HTTP port (port 80) was not allowed in Security Group. The traffic is blocked.
- Open port 80 to allow the traffic





# MRC Workshop – Security Group

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- Setting up security groups
  - Security groups act as a virtual firewall that controls the traffic for one or more instances
  - It contains a set of security rules
  - Default security group only allows SSH (from anywhere)
  - Create a security group
  - Create a rule
    - CIDR ([https://en.wikipedia.org/wiki/Classless\\_Inter-Domain\\_Routing](https://en.wikipedia.org/wiki/Classless_Inter-Domain_Routing))
      - 128.250.0.0/16 (All IPs from UoM)
      - x.x.x.x/32 (x.x.x.x only, use CIDR calculate for IP range calculation)
    - Security Groups
  - Attach the security group

**!! Important !!** Only open the ports that are needed! Allow access from the trusted source!



# MRC Workshop – Volume Storage

- Create a volume
  - **!! Important !!** Must be in the same availability zone as the instance, i.e. **melbourne-qh2-uom**
  - Must within your resource quota.

Create Volume

Volume Name

demo1

Description

Volume Source

No source, empty volume

Size (GiB) \*

10

Availability Zone

melbourne-qh2-uom

Group ?

No group

Type

performance

Description:

Volumes are block devices that can be attached to instances.

Volume Type Description:

Volume Limits

Total Gibibytes

2,159 of 7,168 GiB Used

Number of Volumes

9 of 7,168 Used

Cancel

Create Volume



# MRC Workshop – Volume Storage

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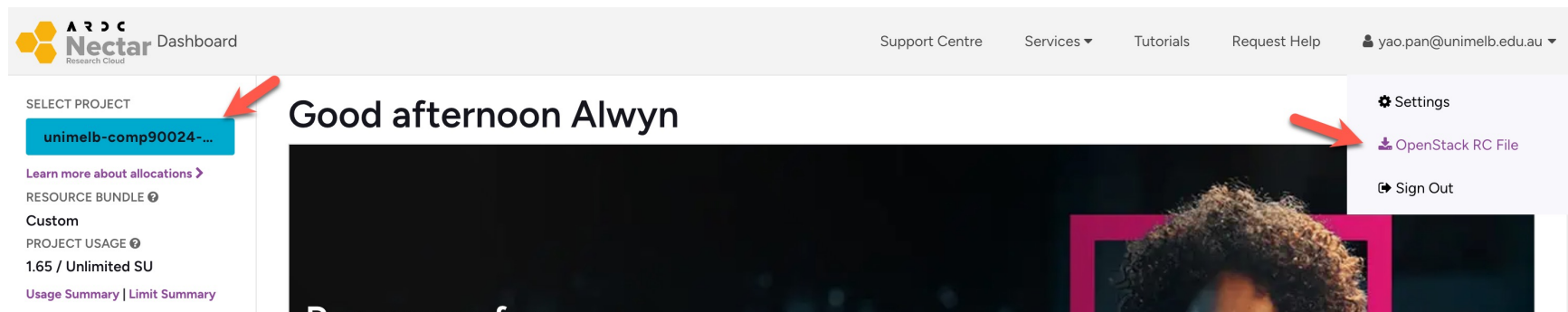
- Attach the volume to an instance
- Mount the volume
  - Check the device name: *sudo fdisk -l*
  - Format the volume: *sudo mkfs.ext4 /dev/vdb*
  - Create the mounting point: *sudo mkdir /mnt/demo*
  - Mount the volume: *sudo mount /dev/vdb /mnt/demo*
  - Check the result: *df -h*

**Note:** if you don't know Linux or what commands like *df -h* do, then Google is your friend! See later too...



# MRC Workshop – OpenRC file

- Prerequisites:
  1. Login to <https://dashboard.rc.nectar.org.au/>
  2. Download *openrc.sh* from Dashboard
    - Make sure the correct project is selected
    - Download the OpenStack RC File





# MRC Workshop – API Password

- Prerequisites:

3. Reset API password

Dashboard -> User -> Settings -> Reset Password

The screenshot shows the Nectar Dashboard interface. The top navigation bar includes links for Support Centre, Services, Tutorials, Request Help, and a user profile dropdown for yao.pan@unimelb.edu.au. The left sidebar contains a 'SELECT PROJECT' section with a button for 'unimelb-comp90024-...', followed by links for 'Learn more about allocations', 'RESOURCE BUNDLE', 'Custom', 'PROJECT USAGE', and 'Usage Summary | Limit Summary'. Below this is a 'Home' section with a list of links: Project, Identity, Settings (expanded), User Settings, Reset Password (highlighted with a red arrow and number 3), My Details, Applications, and Allocations. The main content area displays the 'Password Reset Form' with a description: 'To access the NeCTAR Cloud using the OpenStack API you will need to generate a password. By doing so **any existing password will be forgotten**. To generate a new password click "Reset Password" button.' A red arrow and number 4 point to the 'Reset Password' button. In the top right corner, a red arrow and number 1 point to the user profile dropdown, and a red arrow and number 2 point to the 'Settings' option within the dropdown menu.





# MRC Workshop – macOS vs Windows

- For Mac and Linux Users
  - Congratulations! You are all set.
- For Windows Users
  - Install WSL: [instructions](#).
  - Use Ubuntu as your Linux Distribution.

```
alwynpan@KAMRUI: ~$ screenfetch -p
```

```
./+o+~  
yyyyy- ~yyyyyy+  
://+/////~-yyyyyyo  
.++ ://+++++/~.+sss/^  
.:+to: /+++++++/:--:/-  
o:+o+:+. `.`.'`'-/oo+++++/  
.:+o:+o/.      `+sssoo+/  
..++/+:+oo+o:`    /sssooo.  
/+++//+:`'oo+o     /:-:..  
\t/+o+++`o+o+o      ++//..  
.++.o+++oo+:`       /dddhhh.  
..+.o+oo:.          `oddhhhh+  
 \+.++o+o`'-~~~~~.:ohdhhhhh+  
   .o+++ `ohhhhhhhhyo++os:  
   .o:`.syhhhhhhh/.oo+o`  
         /osyyyyyyo++ooo++/  
        ^^^^^^ +oooo+o\:  
              `oo++.
```

alwynpan@KAMRUI

OS: Ubuntu 22.04 jammy(on the Windows Subsystem for Linux)

Kernel: x86\_64 Linux 5.15.146.1-microsoft-standard-WSL2

Uptime: 3d 10h 28m

Packages: 496

Shell: bash 5.1.16

Resolution: No X Server

WM: Not Found

GTK Theme: [GTK3]

Disk: 286G / 4.8T (7%)

CPU: Intel N100 @ 4x 806MHz

RAM: 1420MiB / 7834MiB



# MRC Workshop – Install OpenStack Client

---

- Follow [this instruction](#) to install OSC
  - It **SUPPORTS** Python3!!!
  - Install pip: *sudo apt install python3-pip* (Linux only)
  - Install OpenStack Client: *sudo pip install python-openstackclient*
  - Install the following OS Project Clients
    - *sudo pip install python-cinderclient*
    - *sudo pip install python-glanceclient*
    - *sudo pip install python-magnumclient*
    - *sudo pip install python-neutronclient*
    - *sudo pip install python-novaclient*
    - *sudo pip install python-swiftclient*



# MRC Workshop – Create instance via OSC

- Source the OpenRC file

Export the environment variables required to interact with OpenStack API

*source ./<your openrc file>*

- Create an instance via OpenStack CLi

*openstack server create --image 49f677df-01e5-45c9-9611-609ef21f60e1 --flavor uom.general.1c4g --network f0c86d08-d45b-45c4-9216-b8abd6bc133c --key-name id\_alwyn --availability-zone melbourne-qh2-uom demo2*

| Field                                | Value                                                                        |
|--------------------------------------|------------------------------------------------------------------------------|
| OS-DCF:diskConfig                    | MANUAL                                                                       |
| OS-EXT-AZ:availability_zone          |                                                                              |
| OS-EXT-STS:power_state               | NOSTATE                                                                      |
| OS-EXT-STS:task_state                | scheduling                                                                   |
| OS-EXT-STS:vm_state                  | building                                                                     |
| OS-SRV-USG:launched_at               | None                                                                         |
| OS-SRV-USG:terminated_at             | None                                                                         |
| accessIPv4                           |                                                                              |
| accessIPv6                           |                                                                              |
| addresses                            |                                                                              |
| adminPass                            | pUstT4bdTFPh                                                                 |
| config_drive                         |                                                                              |
| created                              | 2024-03-17T13:30:12Z                                                         |
| flavor                               | uom.general.1c4g (5e433981-e5d4-4175-b24b-b804d5252199)                      |
| hostId                               |                                                                              |
| id                                   | 8b3767e7-fde0-455d-b7bd-9e611e24f62e                                         |
| image                                | NeCTAR Ubuntu 22.04 LTS (Jammy) amd64 (f3f2983a-3159-441a-ac9d-02aa7a9b39be) |
| key_name                             | mykeypair                                                                    |
| name                                 | demo2                                                                        |
| os-extended-volumes:volumes_attached | []                                                                           |
| progress                             | 0                                                                            |
| project_id                           | 78d23ce7c44940188adecall08613b75                                             |
| properties                           |                                                                              |
| security_groups                      | name='default'                                                               |
| status                               | BUILD                                                                        |
| updated                              | 2024-03-17T13:30:12Z                                                         |
| user_id                              | 49d842414ad345558882e2ad63d60769                                             |

**Note:** You don't need to worry about the commands. All commands will be provided in later workshops ...



# MRC Workshop – Create instance via OSC

---

- Create a volume via OpenStack CLI  
*openstack volume create --size 10 --type performance --availability-zone melbourne-qh2-uom **demo2-volume***
- Attach a volume to an instance via OpenStack CLI  
*openstack server add volume **demo2 demo2-volume***
- Create a security group via OpenStack CLI  
*openstack security group create **demo2-sg***
- Create a security group rule via OpenStack CLI  
*openstack security group rule create --protocol tcp --dst-port 80 --remote-ip 0.0.0.0/0 **demo2-sg***
- Attach a security group to an instance via OSC  
*openstack server add security group **demo2 demo2-sg***



# Basic recipes to follow...

---

- SSH

- *ssh-keygen -t rsa -f <your keypair name>*
- *chmod 600 <your private key>* (or get **UNPROTECTED PRIVATE KEY FILE!** error)
- *ssh -i <your private key> ubuntu@<instance\_ip>* (Ubuntu)

- Volume

- *sudo fdisk -l*
  - *sudo* = runs commands with security privileges of another user (by default: superuser)
- *sudo mkdir /mnt/demo*
- *sudo mkfs.ext4 /dev/vdb*
  - *mkfs* = make file system
  - *ext4* = type of file system (ext2, ext3, ext4, etc)
    - See <http://www.thegeekstuff.com/2011/05/ext2-ext3-ext4/> for details on differences
- *sudo mount /dev/vdb /mnt/demo*
- *df -h*



# Basic recipes to follow...

---

- Install software
  - *sudo apt-get install vim*
    - *apt-get* installs/removes packages on Ubuntu installations
- Common Shell Commands
  - <http://www.dummies.com/computers/operating-systems/linux/common-linux-commands/>